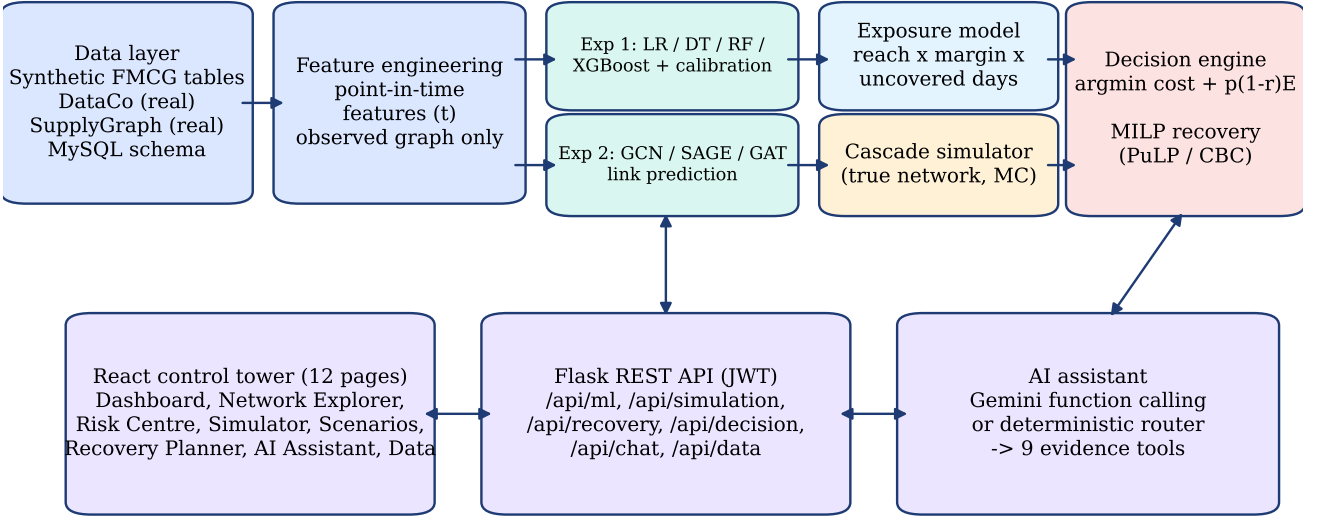Supply Chain Black Box - system architecture